

3.2 $\langle f, g \rangle = \int_1^5 f(x)g(x)dx$; (w_1, w_2, w_3) orthonormal base

$$a) \vec{w}_1 = \frac{1}{\|\vec{1}\|} = \frac{1}{\sqrt{\int_1^5 dx}} = \frac{1}{\sqrt{4}} = \frac{1}{2}$$

$$\vec{w}_2 = \left\langle \frac{1}{2}; x \right\rangle \cdot \frac{1}{2} = \frac{1}{2} \int_1^5 \frac{1}{2} x dx = \frac{1}{2} \cdot 6 = 3$$

$$\vec{w}_2 = \frac{x-3}{\|x-3\|} = \frac{x-3}{\sqrt{\int_1^5 (x-3)^2 dx}} = \frac{x-3}{\sqrt{\frac{125}{3} - 75 + 45 - \frac{1}{3} + 3 - 9}} = \frac{x-3}{\sqrt{\frac{16}{3}}} = \frac{\sqrt{3}(x-3)}{4} \approx 0,43x + 1,29$$

$$\vec{w}_3 = \left\langle \frac{1}{2}; x^2 \right\rangle \cdot \frac{1}{2} + \left\langle \frac{\sqrt{3}(x-3)}{4}; x^2 \right\rangle \cdot \frac{\sqrt{3}(x-3)}{4} = \int_1^5 \frac{1}{2} x^2 dx \cdot \frac{1}{2} + \frac{\sqrt{3}(x-3)}{4} \int_1^5 x^2 \cdot \frac{\sqrt{3}(x-3)}{4} dx =$$

$$= \frac{62}{6} + 6(x-3)$$

$$w_3 = \frac{x^2 - \frac{31}{3} + 6(x-3)}{\sqrt{\int_1^5 \left(x^2 - \frac{31}{3} + 6(x-3)\right)^2 dx}} = \frac{x^2 - \frac{31}{3} - 6(x-3)}{16\sqrt{5}} \approx 0,42x^2 - 2,52x + 3,21$$

$$B(w_1, w_2, w_3) = \left(\frac{1}{2}; 0,43x + 1,29; 0,42x^2 - 2,52x + 3,21 \right)$$

b) $\vec{v}$ = ortogonalni projekce $\sin x$ na V

$$\vec{v} = \langle \vec{w}_1; \sin x \rangle \vec{w}_1 + \langle \vec{w}_2; \sin x \rangle \vec{w}_2 + \langle \vec{w}_3; \sin x \rangle \vec{w}_3 = \frac{1}{2} \int_1^5 \sin \frac{\sin x}{2} dx + \int_1^5 \sin x \frac{\sqrt{3}(x-3)}{4} dx +$$

$$+ \frac{\sqrt{3}(x-3)}{4} + \frac{x^2 - \frac{21}{3} - 6(x-3)}{16\sqrt{5}} \int_1^5 \sin x \cdot \frac{x^2 - \frac{21}{3} - 6(x-3)}{16\sqrt{5}} dx = 0,064 \approx 1,49(0,43x - 1,299) -$$

$$- 0,13(0,42x^2 - 2,52x + 3,21) = \underline{\underline{-0,05x^2 - 0,31x + 1,58}}$$

c) $\|\sin x + 0,05x^2 + 0,31x - 1,58\| = \sqrt{\int_1^5 (\sin x + 0,05x^2 + 0,31x - 1,58)^2 dx} = \underline{\underline{0,558}}$